HW 7

1

For the three functions f_1, f_2, f_3, design multiple-output network that minimize the number of gates and connections. Derive and INV-NOR-NOR-INV PLA for your resulting circuit.

$f_1 = wxy + \bar{w}\bar{y}z$
$f_2 = \bar{x}\bar{z} + wxz + x\bar{y}z$
$f_3 = \bar{x}\bar{z} + wxyz + \bar{w}\bar{x}\bar{y}$

✓ **Answer**

1s	Binary	Function
0	0000.	2, 3
1	0001	1, 3
1	0010.	2, 3
1	1000.	2, 3
2	0101	1, 2
2	1010.	2, 3
3	1110.	1
3	1101.	2
4	1111	1, 2, 3

1s	Binary	Function
0	000_	3
0	00_0.	2, 3
0	_000.	2, 3
1	0_01	1
1	10_0.	2, 3
1	_010.	2, 3
2	_101	2
3	111_	1
3	11_1	2

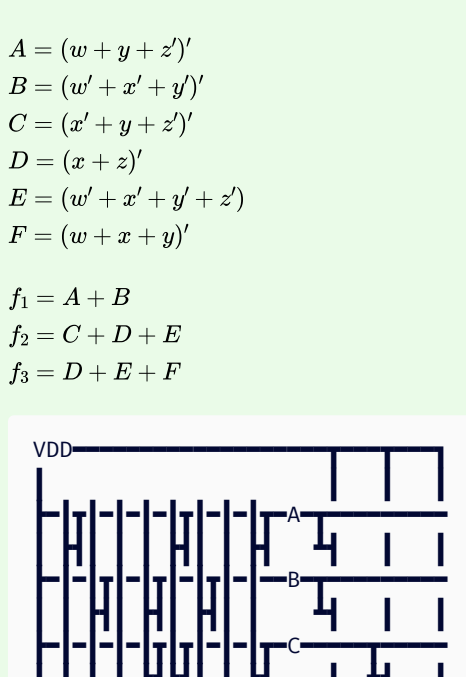
1s	Binary	Function
0	_0_0	2, 3

	Functions	$w'x'y'z$ 0001	$w'xy'z$ 0101	$w'x'y'$ 000_	$w'y'z$ 0_01	$xy'z$ _101	wxz 11_1	$wxyz$ 1111	wxy 111_	$x'z'$ _0_0
Functions		1, 3	1, 2	3	1	2	2	1,2,3	1	2, 3
0000	2, 3			e						e
0001	1, 3	oo		O	o					
0101	1, 2		uo		O	O				
1101	2					o	u			
1111	1, 2, 3						e	eeE	e	
0010	2, 3									E
1000	2, 3									E
1010	2, 3									E
1110	1								E	

$f_1 = w'y'z + wxy$
$f_2 = xy'z + x'z' + wxyz$
$f_3 = w'x'y' + wxyz + x'z'$

$A = (w + y + z')'$
$B = (w' + x' + y')'$
$C = (x' + y + z')'$
$D = (x + z)'$
$E = (w' + x' + y' + z')$
$F = (w + x + y)'$

$f_1 = A + B$
$f_2 = C + D + E$
$f_3 = D + E + F$



2

Apply the first Tison method to derive the set of all prime implicants for each of the following functions:

a

$f = vxz + wxz' + ux'yz + wx'yz + v'yz'$

✓ **Answer**

Consensus Terms:
$uvyz + vwyz + vwx + uv'x'y + v'wx'y$

No reduction possible,
$f = vxz + wxz' + ux'yz + wx'yz + v'yz' + uvyz + vwyz + vwx + uv'x'y + v'wx'y$

b

$f = twy + t'v'w + uvw + uvy' + vxy' + vxz + uv'wy + v'wyz$

✓ **Answer**

t	u	v	w	x	y	z
1			1		1	
-1		-1	1			
	1	1	1			
	1	1			-1	
		1		1	-1	
		1		1		1
	1	-1	1		1	
		-1	1		1	1

Consensus

t	u	v	w	x	y	z
		-1	1		1	
-1	1		1			
-1	1		1		-1	
-1			1	1	-1	
-1			1	1		1
	1		1		1	
	1		1	1	1	1
	1		1		1	1
			1	1	1	1

Reduction

t	u	v	w	x	y	z
1			1		1	
-1		-1	1			
	1	1	1			
	1	1			-1	
		1		1	-1	
		1		1		1
		-1	1		1	
-1	1		1			
	1		1		1	
-1			1	1	-1	
-1			1	1		1
			1	1	1	1

Consensus

t	u	v	w	x	y	z
1		1	1	1		
		1	1	1		1
-1		-1	1	1		
-1	1		1	1		
-1			1	1		1

Reduction

t	u	v	w	x	y	z
-1	1		1			
-1			1	1		1
-1		-1	1			
	1	1	1			
		1		1		1
1		1	1	1		
-1			1	1	-1	
	1	1			-1	
		1		1	-1	
	1		1		1	
			1	1	1	1
1			1		1	
		-1	1		1	

No new Consensus

$f = t'uw + t'wxz + t'v'w + uvw + vxz$
$+tvwx + t'wxy' + uvy' + vxy'$
$+uwy + wxyz + twy + v'wy$

3

Apply the second Tison method to derive the set of all irredundant sum for each of the following list of prime implicants:

a

Indexed PI's: $x'y'A, x'z'B, y'zC, xzD$

✓ **Answer**

minterm	PIs covered
000	A, B
001	A, C
010	B
011	
100	
101	C, D
110	
111	D

$f = (A + B)(A + C)(B)(C + D)D$
$= BD(A + C)$

The irredundant sums are
$A + B + D$ and $B + C + D$

b

✓ **Answer**

Indexed PI's: $wxzA, xyzB, w'yzC, vw'xy'D, vw'xz'E, w'xy'z'F, v'w'x'zG, v'w'yz'H, v'w'x'y'I$

Check for essential PIs:

$A(x' + y' + z') = wxy'z$
$*(w + y' + z') = wxy'z$
$*(v' + w + x' + y) = wxy'z$
$*(v' + w + x' + z) = wxy'z$
$*(w + x' + y + z) = wxy'z$
$*(v + w + x + z') = wxy'z$
$*(v + w + y' + z) = wxy'z$
$*(v + w + x + y) = wxy'z$
A is essential

$B * A' \rightarrow w'xyz$
$*C' \rightarrow 0$
B is nonessential

$C * A' * B' \rightarrow w'x'yz$
$*D' * E' * F' * G' \rightarrow vw'x'yz$
$*H' * I' \rightarrow vw'x'yz$
C is essential

$D * A' * B' * C' * E' \rightarrow vw'xy'z$
$*F' \rightarrow 0$
D is nonessential

$E * A' * B' * C' * D' \rightarrow vw'xy'z'$
$*F' \rightarrow 0$
E is nonessential

$F * A' * B' * C' * D' \rightarrow vw'xy'z'$
$*E' \rightarrow 0$
F is nonessential

$G * A' * B' \rightarrow v'w'x'y'z$
$*C' * D' * E' * F' * H' * I' \rightarrow v'w'x'y'z$
G is essential

$H * A' * B' * C' * D' * E' * F' * G' * I' \rightarrow v'w'xyz'$
H is essential

$I * A' * B' * C' * D' * E' * F' * G' * H' \rightarrow v'w'x'y'z'$
I is essential

Check for remaining prime implicants

$BA'C'G'H'I' \rightarrow 0$
B is covered

$DA'C'G'H'I' \rightarrow (w'x)vy'$
D is uncovered

$EA'C'G'H'I' \rightarrow (w'x)vz'$
E is uncovered

$FA'C'G'H'I' \rightarrow (w'x)y'z'$
F is uncovered

DEF is circular, we need two of the 3 to cover the function.

$f = ACGHI(DE + DF + EF)$

4

For the circuit shown:

a

Determine the 1-sets.

✓ **Answer**

$A = (a'b')'$
$B = (ad)'$
$C = (a'cd')'$
$D = (ABc)'$
$f = (CD)'$

$f = ((a'cd')'(ABc)')'$
$f = a'cd' + ABc$
$f = a'cd' + (a + b)(a' + d')c$
$f = a'cd' + aa'c + acd' + a'bc + bcd'$

b

Analyze the circuit for static hazards using the 1-sets.

✓ **Answer**

$aa'c$ poses a hazard.

The consensus of $a'cd', acd'$ is cd, and cd is not present, so this also poses a hazard.

There are no other missing consensi.

c

Can you derive a procedure to detect static hazards in combinational logic circuits using product of sum representation?

✓ **Answer**

Determine all consensus terms of the 1- or 0- sets, and if any of those consensus terms are missing or unabsorbable completely by a singular term, then a transition hazard may exist.

5

Given an indexed list of $PI = w'zA, xy'B, yzC, xzD, wxE, w'x'yF$ for a function $f(w, x, y, z)$: Use the second Tison method with variables in the w, x, y, z to find the petrick expression.

✓ **Answer**

minterm	PIs	Contains Essential
0000		
0001	A	1
0010	F	1
0011	A,C,F	1
0100	B	1
0101	A,B,D	1
0110		
0111	A,C,D	1
1000		
1001		
1010		
1011	C	1
1100	B,E	1
1101	B,D,E	1
1110	E	1
1111	C,D,E	1

A, B, D, E, F are essential and there are no uncovered terms.

$f = ABDEF$